SI 608 – Networks

School of Information, University of Michigan, Fall 2025
Class sessions: Thursdays 9-12, 2290 LCSIB

Instructor: Ceren Budak (cbudak@umich.edu)
Office hours: Thursdays 1-3pm @ Leinweber 5120.
Office hours on October 9 are canceled due to Prof Budak being out of town.

GSI: Laura Kurek (lkurek@umich.edu)

- Office hours: Wednesdays 9-11am @ Leinweber 5460/5555
    - Office hours will be canceled Oct 22 due to conference travel. I will hold the following make-up sessions:
        - Monday Oct 27 10-11am
        - Tuesday Oct 28 10-11am
    - For virtual office hours, please send an email to schedule a virtual appointment. Use the following link to join:
        - https://umich.zoom.us/j/95015963785?jst=2
        - Meeting ID: 950 1596 3785

Instructional Aides: Ayush Shah (shahayu@umich.edu), Shaoying Zheng (shaoyinz@umich.edu), Huanyu Ren (koberen@umich.edu)

- Office hours (with Ayush): Mondays 4-6pm @ Leinweber 5460

Note: Some syllabus details are subject to change.

Course Overview

We will cover topics in network analysis in this course, including social networks and information networks (e.g., the web). We will introduce basic concepts in network theory, metrics to characterize networks, models to explain the generation of networks, and methods to further analyze networks. In the lab sessions, the students will experience various analysis tools to analyze real-world network data. In the second half of the course, we will introduce a wide variety of applications of network analysis to real-world problems such as information retrieval. The final course project will provide students the opportunity to apply the concepts and techniques learned in class to networks of their interest. Students will also participate in the class through group discussions of related papers.

Learning Objectives

Upon completion of this course, students should be able to:

- Understand basic concepts and metrics in network analysis
- Understand how to characterize, model, and analyze a network
- Utilize a variety of tools to manage, measure, visualize, and analyze networks
- Apply concepts and techniques of network analysis to real-world problems involving networks
- Start working toward becoming a professional with expertise in network analysis

Course Evaluation

- **Paper Presentations/Readings (15%):** We will have a set of papers to discuss each week (except for the first 2 and the last weeks). Students will form teams of two (or more depending on class size) and present a paper. The goal of student paper presentations is: (i.) to delve deeper into the concepts covered in class the previous week and (ii.) to practice and improve our presentation skills. Your grade is based on your presentation as well as your engagement with others' presentations/papers. 8% will be based on your engagement and 7% will be based on your engagement with other papers across the semester. Your engagement with the papers throughout the semester will be decided based on your engagement on Perusall assignments.
- **Four homework assignments (30%):** No late submissions will be accepted under *any* normal circumstances. Extensions will only be granted under extreme circumstances to students with *documented* reasons (e.g. medical) at the instructor's discretion.
- **Midterm Exam (15%):** The midterm exam will be in person.
- **Course Project (40%):** The projects will be a group effort. Students need to form groups (3-4 people) and define their project in the first few weeks of the class.
    - Proposal: 4%
    - Midterm Evaluation: 8%
    - Final presentation: 8%
    - Final report: 20%

Late homework policy: The homework assignments are due when the next class begins. No late submissions will be accepted under *any* normal circumstances. Extensions will only be granted under extreme circumstances to students with *documented* reasons (e.g. medical) at the instructor's discretion.

Lab submission policy: Labs are not submitted or graded.

Class format

Typical format will be as follows:

- Group Discussions about papers (except for week 1, 2, and 14): ~30 mins
- Lecture: ~90 mins
- Lab/Project: ~60mins

Text

- David Easley and Jon Kleinberg. Networks, Crowds, and Markets. To be published by Cambridge University Press, Spring 2010. (**EK2010**)
- Newman, M. E. J. "The Structure and Function of Complex Networks." SIAM Review. 45 (2003): 167-256. (**N2003)**
- (Optional) Wouter de Nooy, Andrej Mrvar, and Vladimir Batagelj. Exploratory Social Network Analysis with Pajek. Structural analysis in the social sciences. New York: Cambridge University Press, 2005. (Pajek)

Classroom Policy

Lectures will be held in person but will also be made available online. The lectures will be available asynchronously through Lecture Capture. Our guidance is to provide residential instruction this year. As such, the lectures are geared toward this form of instruction. Please share your feedback/struggles, so we can try to find ways to improve your educational experience.

Students are asked to attend class on time and remain (in person) through the entire class. I don't take attendance in this class, but we do have in-class labs that must be completed for credit. These labs do not need to be completed during lecture time--you will have 1 week to turn in the labs. But the instructional team will be available during class time to help guide you if you have questions. If attending the class in person, please wear your mask throughout the class.

Communication

The best way to contact any member of the teaching team is (i.) attending the lectures and asking your questions there or (ii.) by using Piazza. We will try to answer questions that are sent via Piazza within about 48 hours. Responses on weekends and holidays may be slower. If you have questions about course material, homework, or labs, please attend office hours. Your GSI should be your first choice for technical questions; conceptual questions are best directed to Dr. Budak.

Personal matters should be communicated via email to Dr. Budak. Please include “[SI 618]” in your subject line to receive a timely response.

Academic Integrity and Misconduct

Unless otherwise specified in an assignment all submitted work must be your own, original work. Any excerpts, statements, or phrases from the work of others must be clearly identified as a quotation, and a proper citation provided. The instructors actively check for plagiarism and have zero tolerance. If caught, you will receive a zero for that portion of the assignment (or the whole assignment), depending on the scope of the plagiarism. All cases will be referred to the Office of Academic Integrity, regardless of their scope. Please (please) do not plagiarize; we hate finding these cases. If you are stressed and tempted to submit other’s work, please come talk to us first.

Re-using worked examples of significant portions of your own code from another class or code from other people from places like online tutorials, Kaggle examples, or Stack Overflow will be treated as plagiarism. If you are in doubt, please contact one of the instructors to check.

Any violation of the University's policies on Academic and Professional Integrity may result in serious penalties, which might range from failing an assignment, to failing a course, to being expelled from the program. Violations of academic and professional integrity will be reported to Student Affairs. Consequences impacting assignment or course grades are determined by the faculty instructor; additional sanctions may be imposed.

ChatGPT / CoPilot / Gen AI Policy:

<This policy is subject to change before the semester starts. Please attend the first lecture for the most up-to-date policy.>

This course material is designed to have you generate solutions and further your skill. It is ok to use genAI as a tool to further your understanding but it is strictly forbidden to use them to provide answers to assignments or other assessment. You continue to learn while working on homeworks so you might reach a point while working on your assignments where you rely on these tools for clarification or conceptual questions. That is acceptable. Here is the bottomline: You need to 1) not use genAI to generate the answer (or parts of the answer) but ask clarification and conceptual questions, 2) turn in the set of prompts you used for each assignment (please make sure to include the entire related history).

Furthermore, note that we will have in class components where a subset of the questions and your answers will be presented to you in quiz form and only in cases where students can explain their answers will their homeworks receive full grade. If you fail to answer these questions repeatedly, it will result in a failing grade.

Collaboration: UMSI strongly encourages collaboration while working on some assignments, such as homework problems and interpreting reading assignments as a general practice. Active learning is effective. Collaboration with other students in the course will be especially valuable in summarizing the reading materials and picking out the key concepts. You must, however, write your homework submission on your own, in your own words, before turning it in. If you worked with someone on the homework before writing it, you must list any and all collaborators on your written submission. Each course and each instructor may place restrictions on collaboration for any or all assignments. Read the instructions careful and request clarification about collaboration when in doubt. Collaboration is almost always forbidden for take-home and in class exams.

Plagiarism: All written submissions must be your own, original work. Original work for narrative questions is not mere paraphrasing of someone else's completed answer: you must not share written answers with each other at all. At most, you should be working from notes you took while participating in a study session. Largely duplicate copies of the same assignment will receive an equal division of the total point score from the one piece of work. You may incorporate selected excerpts, statements or phrases from publications by other authors, but they must be clearly marked as quotations and must be attributed. If you build on the ideas of prior authors, you must cite their work. You may obtain copy editing assistance, and you may discuss your ideas with others, but all substantive writing and ideas must be your own, or be explicitly attributed to another. See the relevant student handbooks available on the UMSI intranet or the CSE / Engineering honor code for the definition of plagiarism (e.g., (Doctoral, Rackham, MSI, MHI, or BSI), resources to help you avoid it, and the consequences for intentional or unintentional plagiarism.

Accommodations for Students with Disabilities

The University of Michigan recognizes disability as an integral part of diversity and is committed to creating an inclusive and equitable educational environment for students with disabilities. Students who are experiencing a disability-related barrier should contact [Services for Students with Disabilities](https://ssd.umich.edu/) https://ssd.umich.edu/; 734-763-3000 or ssdoffice@umich.edu). For students who are connected with SSD, accommodation requests can be made in Accommodate. If you have any questions or concerns please contact your SSD Coordinator or visit SSD's Current Student webpage. SSD considers aspects of the course design, course learning objects and the individual academic and course barriers experienced by the student. Further conversation with SSD, instructors, and the student may be warranted to ensure an accessible course experience. The instructional team will treat any information that you provide in as confidential a manner as possible.

Student Mental Health and Wellbeing

Students may experience stressors that can impact both their academic experience and their personal well-being. These may include academic pressure and challenges associated with relationships, mental health, alcohol or other drugs, identities, finances, etc.
If you are experiencing concerns, seeking help is a courageous thing to do for yourself and those who care about you. If the source of your stressors is academic, please contact me so that we can find solutions together. For personal concerns, U-M offers the following resources:

- Counseling and Psychological Services (CAPS) - confidential; 734-764-8312; for after-hours urgent support, call and press 0; counseling, workshops, groups and more. Ashley Evearitt, a CAPS counselor is embedded in UMSI, information about how to schedule an appointment with her can be found here.
- Dean of Students Office - 734-764-7420; provides support services to students and manages critical incidents impacting students and the campus community
- Ginsberg Center for Community Service Learning - 734-763-3548; opportunities to engage as learners and leaders to create a better community and world
- Maize and Blue Cupboard (MBC) - 734-936-2794; Food pantry with groceries, kitchen and cooking supplies, personal and household items, and support
- Multi-ethnic Student Affairs (MESA) - 734-763-9044; diversity and social justice through the lens of race and ethnicity
- Office of Student Conflict Resolution - 734-936-6308; offers multiple pathways for resolving conflict
- Office of the Ombuds - 734-763-3545; students can raise questions and concerns about the functioning of the university.
- Services for Students with Disabilities (SSD) - 734-763-3000; accommodations and access to students with disabilities
- SilverCloud - Students may also use an online, self-guided, interactive mental health resource that provides cognitive behavioral interventions.
- Sexual Assault Prevention and Awareness Center (SAPAC) - confidential; 734-764-7771 or 24-hour crisis line 734-936-3333; addresses sexual assault, intimate partner violence, sexual harassment, and stalking
- Spectrum Center - 734-763-4186; support services for LGBTQ+ students
- Trotter Multicultural Center - 734-763-3670; intercultural engagement and inclusive leadership education initiatives

- University Health Service (UHS) - 734-764-8320; clinical services include nurse advice by phone, day or night
- Well-being for U-M Students website - searchable list of many more campus resources
- Wolverine Wellness - 734-763-1320; provides Wellness Coaching, Collegiate Recovery Program, and much more

Schedule (Tentative)

- **Week 1: Course Overview; Introduction to Networks**
- **Week 2: Basic Concepts. Network Metrics I**
  - *Reading material:*
    - EK2010 Chapter 1&2
    - N2003 Section 1-3
  - *Lab:*Introduction to NetworkX
- **Week 3: Centrality. Network Metrics II**
  - Papers to be discussed
    - "What is twitter, a social network or a news media?" Kwak et al. 2010
    - "Measurement and analysis of online social networks." Mislove et al. 2007
    - "Analysis of topological characteristics of huge online social networking services." Ahn et al. 2007
    - "User interactions in social networks and their implications." Wilson et al. 2009
  - *Reading material:*
    - EK2010 Chapter 2,3&14 (introductory material from Chapter 14)
    - Baker, W. E., and R. R. Faulkner. "The Social Organization of Conspiracy: Illegal Networks in the Heavy Electrical Equipment Industry." American Sociological Review. 58. 6 (1993): 837.
    - Friedkin, N. E. "Structural Bases of Interpersonal Influence in Groups: A Longitudinal Case Study." American Sociological Review. 58. 6 (1993): 861. (JSTOR account required for article)
    - Optional: Wilson, Christo, Bryce Boe, Alessandra Sala, Krishna PN Puttaswamy, and Ben Y. Zhao. "User interactions in social networks and their implications." In *Proceedings of the 4th ACM European conference on Computer systems*, pp. 205-218. Acm, 2009.
    - Optional: Chapter 5: Centrality and prestige - Wasserman, Stanley, and Katherine Faust. Social Network Analysis: Methods and Applications. Structural analysis in the social sciences, 8. Cambridge: Cambridge University Press, 1994.

        - Optional: Pajek Chapter 6: Center and Periphery
        - Optional: Pajek Chapter 9: Prestige
    - *Lab:*Centrality in NetworkX
- **Week 4: Small-world Networks. Network Models I**
    - Papers to be discussed
        - “Structural Bases of Interpersonal Influence in Groups: A Longitudinal Case Study.” By Friedkin, N. E.
        - "The Social Organization of Conspiracy: Illegal Networks in the Heavy Electrical Equipment Industry." Baker, W. E., and R. R. Faulkner.
        - Book: "#HashtagActivism" Chapter: "Racial Violence and Racial Profiling: From #OscarGrant to #TrayvonMartin." Sarah Jackson, Moya Bailey, Brooke Foucoult Welles.
        - "Interdisciplinarity, Gender Diversity, and Network Structure Predict the Centrality of AI Organizations." Madalina Vlasceanu et al. https://dl.acm.org/doi/abs/10.1145/3531146.3533069Links to an external site.
        - “The role of women scholars in the Chilean collaborative educational research: a social network analysis.” https://www.jstor.org/stable/45199891. Vol. 78, No. 1 (July 2019), pp. 115-131
        -
    - *Reading material:*
        - EK2010 Chapter 20: Small World Phenomenon
        - N2003 Section 6
        - Watts DJ, and SH Strogatz. "Collective Dynamics of 'small-World' Networks." Nature. 393. 6684 (1998): 440-2. (Nature subscription required for article)
        - Travers, Jeffrey & Stanley Milgram. 1969. "An Experimental Study of the Small World Problem." Sociometry, Vol. 32, No. 4, pp. 425-443.
        - Dodds, Peter Sheridan, Roby Muhamad, and Duncan J. Watts. "An experimental study of search in global social networks." *Science*5634 (2003): 827-829.
    - *Lab:* Small World Networks
    - HW 1 released
- **Week 5: Scale-free Networks. Network Models II**
    - Papers to be discussed
        - “Identity and Search in Social Networks” by Duncan J. Watts, Peter Sheridan Dodds, and M. E. J. Newman
        - “Four degrees of separation” by Lars Backstrom et al.
    - Reading material:

        - "Power laws, Pareto distributions and Zipf's law" M. E. J. Newman
        - "Topology of Evolving Networks: Local Events and Universality" Réka Albert and Albert-László Barabási
        - "Emergence of Scaling in Random Networks" Albert-Laszlo Barabasi and Reka Albert
        - EK Chapter 18
    - Lab: Scale Free Networks
    - HW1 due
- **Week 6: Networks over Time**
    - Papers to be discussed
        - "Scale-free networks are rare" by Broido and Clauset
        - "Scale-free is not rare in international trade networks" by Linqing Liu, Mengyun Shen and Chang Tan. https://www.nature.com/articles/s41598-021-92764-1.
        - "Classification of Scale-free networks" by Goh et al. https://www.jstor.org/stable/3073266?seq=6
    - *Reading material:*
        - "Empirical Analysis of an Evolving Social Network" Gueorgi Kossinets and Duncan J. Watts
        - "Team Assembly Mechanisms Determine Collaboration Network Structure and Team Performance" Roger Guimerà, Brian Uzzi, Jarrett Spiro, and Luís A. Nunes Amaral
        - "Group Formation in Large Social Networks: Membership, Growth, and Evolution" Lars Backstrom, Dan Huttenlocher, Jon Kleinberg
        - "Graphs over Time: Densification Laws, Shrinking Diameters and Possible Explanations" Jure Leskovec, Jon Kleinberg, Christos Faloutsos
        - *GraphRNN: Generating Realistic Graphs with Deep Auto-regressive Models. J. You, R. Ying, X.Ren, W. L. Hamilton, J. Leskovec. International Conference on Machine Learning (ICML), 2018.*
    - *Lab:* Networks over Time
    - *Project proposals are due
    - *HW 2 assigned
    - Mid-point CRLT course Evaluation: This is not an assignment for you but an opportunity to give feedback about the course during the last 30 mins of the course.
- **Week 7: Communities** – This class will be presented by the GSI.
    - Papers to be discussed
        - "The Anatomy of the Facebook Social Graph" by Ugander et al.

        - "Meeting Strangers and Friends of Friends: How Random are Social Networks?" by Matthew O. Jackson. https://www.aeaweb.org/articles?id=10.1257/aer.97.3.890
        - "Creation and Growth of Online Social Network. How do social networks evolve?" by Musial et al. https://link.springer.com/article/10.1007/s11280-012-0177-1#citeas
    - *Reading material:*
        - Heuristics for community detection, K-means clustering, betweenness clustering, and modularity
        - “Friends and Neighbors on the Web” Adamic, Adar
        - “The Political Blogosphere and the 2004 U.S. Election: Divided They Blog” Adamic, Glance 2005
        - “A Social Network Caught in the Web” Adamic et al.
        - “Finding community structure in very large networks” Clauset et al. 2004
        - “Community structure in social and biological networks” Girvan, Newman 2001
        - “Community Detection and Mining in Social Media” by Tang et al. 2010
    - *HW 2 due
- **Week 8: Ranking, Web**
    - <u>Papers to be discussed</u>
        - “Statistical Properties of Community Structure in Large Social and Information Networks” J. Leskovec, K. Lang, A. Dasgupta, M. Mahoney.
        - “The Political Blogosphere and the 2004 U.S. Election: Divided They Blog” Adamic, Glance 2005
        - “Finding community Structure in Very Large Networks” Clauset et al. 2004
        - “The ground truth about metadata and community detection in networks” Peel et al.
    - *Reading material:*
        - EK2010 Chapter 13&14
        - “LexRank: Graph-based Lexical Centrality as Salience in Text Summarization” Erkan, Radev 2004
    - *Midpoint reviews assigned
- **Week 9: Classification and Recommendation + Midterm Review**
    - <u>Papers to be discussed</u>
        - “The authority of Supreme Court precedent” James H. Fowler and Sangick Jeon 2008
        - “Graph Structure in the Web.” Broder et al. 2000
    - Reading material:
        - “Iterative Classification in Relational Data” Neville, Jensen 2000

        - “Node classification in social networks” Bhagat et al. 2011
        - “Predicting Positive and Negative Links in Online Social Networks” Leskovec et al. 2010
        - “Query Suggestion Using Hitting Time” Mei et al. 2008
        - “Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market” Salganik et al. 2006
    - Project Mid-point review due
    - *HW 3 assigned
- **Week 10: Midterms**
- **Week 11: Diffusion and Cascades**
    - <u>Papers to be discussed</u>
        - “Experimental Study of Inequality and Unpredictability in an Artificial Cultural Market” Salganik et al. 2006
        - “Query Suggestion Using Hitting Time” Mei et al. 2008
    - *Reading material:*
        - “Diffusion of Innovations.” Everett Rogers (this is a book, I don’t expect you to read all of it but it is a nice reference book you should be aware of)
        - “A simple model of global cascades on random networks.” D.J. Watts (2002)
        - “Influentials, Networks, and Public Opinion Formation” D.J. Watts , P.S. Dodds (2007)
        - “Everyone’s an influencer: Quantifying Influence on Twitter” Bakshy et al.
        - EK2010 Chapters 19, 21
    - *HW 3 due
- **Week 12: Influence vs. Homophily**
    - *Papers to be discussed:*
        - “The structure of Virality of Online Diffusion” by Goel et al.
        - “Information diffusion through blogspace” Gruhl, D., Guha, R., Liben-Nowell, D., & Tomkins, A. (2004).
        - "Networks, Information & Social Capital", M. van Alstyne, S. Aral. 2008.
        - "Structure and tie Strengths in mobile Communication networks", Onnela J. et.al. PNAS 2007;104:7332-7336.
        - “The structure of information pathways in a social communication network”, Kossinets et al. KDD 2008
    - *Reading material:*
        - EK2010 Chapter 4: Networks in their surrounding contexts
        - "Maximizing the Spread of Influence Through a Social Network” Kempe et al. 2003
        - "Cost-Effective Outbreak Detection in Networks" Leskovec et al. 2007

        - "Influence and Correlation in Social Networks" Anagnostopoulos et al. 2009
        - “Distinguishing influence-based contagion from Homophily-driven diffusion in dynamic networks” Aral et al. 2009
        - “Randomization tests for distinguishing social influence and homophily effects” La Fond et al. 2010
    - *HW 4 assigned
- **Week 13: Graph Traversal/Search**
    - <u>Papers to be discussed</u>
        - “Origins of Homophily in an Evolving Social Network” by Kossinets et al.
        - "Influence and Correlation in Social Networks" Anagnostopoulos et al. 2009
        - “Randomization tests for distinguishing social influence and homophily effects” La Fond et al. 2010
    - *Reading material:*
        - Breadth-first, depth-first, greedy best-first, and A* search algorithms
        - Sections 3&4 in “Artificial Intelligence: A Modern Approach.” Russell and Norvig (but the material presented in the slides should suffice.)
    - *HW 4 due
- **Week 14: Final Project Presentation**
    - *Project reports due